IMO(2^{nd}) – 1960

First Day

1. Find all the three digit numbers for which one obtains, when dividing the number by 11, the sum of the squares of the digits of the initial number.
2. For which real numbers x does the following inequality hold: $\frac{4x^2}{\left(1-\sqrt{1+2x}\right)^2} < 2x+9$?
3. A right angled triangle ABC is given for which the hypotenuse BC has length a and is divided into n is odd. Let α be the angle with which the point A sees the segment containing the middle of the hypotenuse. Prove that $\tan\alpha = \frac{4nh}{(n^2-1)a}$, where h is the height of the triangle.

Second Day

4. Construct a triangle ABC whose length of height h_a and h_b (from A and B, respectively) and length of median m_a (from A) are given.
5. A cube $ABCDA'B'C'D'$ is given.
   (a) Find the locus of all the midpoints of segments XY, where X is any point on segment AC and Y is any point on segment $B'D'$.
   (b) Find the locus of all points Z on segments XY such that $\overrightarrow{ZY} = 2\overrightarrow{XZ}$.
6. An isosceles trapezoid with bases a, b and height h is given.
   (a) On the line of the symmetry construct the point P such that both (nonbase) sides are seen from P with an angle of 90^0.
   (b) Find the distance of P from one of the bases of the trapezoid.
   (c) Under what conditions for a, b and h can the point P be constructed (analyze all possible cases)?
7. A sphere is inscribed in a regular cone. Around the sphere a cylinder is circumscribed so that its base is in the same plane as the base of the cone. Let V_1 be the volume of the cone and V_2 the volume of the cylinder.
   (a) Prove that $V_1 = V_2$ is impossible.
   (b) Find the smallest k for which $V_1 = kV_2$, and in this case construct the angle at the vertex of the cone.

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